Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 04-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A three-period multicenter study, with a randomized-withdrawal, double-blinded, placebo-controlled design to evaluate the clinical efficacy, safety and tolerability of MAS825 in patients with monogenic IL-18 driven autoinflammatory diseases, including NLRC4-GOF, XIAP deficiency, or CDC42 mutations **Short Title/Acronym:** MASter-1 PoC study **Protocol Number:** CMAS825D12201 **NCT Number:** NCT04641442 **Sponsor Name:** Novartis Pharmaceuticals (Novartis Pharma AG) **Investigational Product:** MAS825 (bispecific anti-IL-1β and IL-18 monoclonal antibody) **Phase of Development:** Phase 2 **Medical Monitor:** Novartis Lead of Regulatory Affairs

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the clinical efficacy, safety, and tolerability of MAS825 in patients with monogenic IL-18 driven autoinflammatory diseases (NLRC4-GOF, XIAP deficiency, or CDC42 mutations). **Secondary Objectives:** To assess the long-term safety, pharmacokinetics, and the prevention of disease flares following a randomized-withdrawal period.

2.2 Background & Rationale Monogenic IL-18 driven autoinflammatory diseases, including NLRC4-gain-of-function (GOF), XIAP deficiency, and CDC42 mutations, result in severe, recurrent, and sometimes life-threatening hyperinflammation and macrophage activation syndrome (MAS). Current therapeutic options are limited and often fail to provide complete disease control. This study is designed to evaluate MAS825, a bispecific monoclonal antibody that simultaneously neutralizes IL-1β and IL-18 pathways, seeking to provide a targeted and effective approach to modulating inflammasome-driven systemic inflammation in these specific genetic populations.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled (during Period 2) **Blinding:** Double-blinded (during Period 2) **Randomization:** 1:1 ratio to MAS825 or matching placebo during the randomized-withdrawal period **Design Description:** This is a three-period study. Period 1 is an open-label, single-arm active treatment. Period 2 utilizes a randomized-withdrawal, double-blinded, placebo-controlled design. Periods 3 and 3s are open-label, long-term safety follow-ups.

3.2 Planned Enrollment & Duration Target N: Rare disease cohort size (not strictly bounded; open to eligible pediatric and adult populations) **Number of Sites:** Multicenter global study (21 locations including sites in the US, Canada, Japan, Europe, and Turkey) **Estimated Study Start:** 18-Dec-2020 **Estimated Primary Completion:** 07-Apr-2032 (Expected overall study completion)

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Genetic diagnosis of either NLRC4-GOF, XIAP deficiency, or CDC42 mutation. 2. Clinical history and investigations consistent with autoinflammation and infantile enterocolitis (AIFEC/NLRC4-GOF), XIAP, or CDC42. 3. For XIAP deficiency, patients must have persistent disease or be resistant to escalating therapy. 4. Male and female patients weighing at least 3 kg.

4.2 Exclusion Criteria 1. History of hypersensitivity to any of the study drugs or to drugs of similar chemical classes. 2. Signs and symptoms of clinically significant active bacterial, fungal, parasitic, or viral infections (e.g., a positive HIV test, positive Hepatitis B surface antigen, positive Hepatitis C, or active Tuberculosis). 3. Pregnant or nursing (lactating) females. 4. Live vaccinations within 1 month prior to MAS825 treatment, during the trial, and up to 3 months following the last dose. 5. Patients weighing > 160 kg at Screening.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: MAS825 administered based on weight/cohort specific parameters. **Route of Administration:** Intravenous (IV) or Subcutaneous (SC) (as determined by the protocol formulation). **Duration of Treatment:** The total study duration across all periods is approximately 4 to 5 years.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care therapies that are not specifically prohibited, with exceptions for stable doses of glucocorticoids or specific therapies per protocol. **Prohibited:** Previous treatment with anti-rejection and/or immunomodulatory drugs within the past 28 days or 5 half-lives (whichever is longer) prior to MAS825 treatment. Live virus vaccines are strictly prohibited.

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6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Prior to Day 0 | Informed Consent, Medical/Genetic History, Labs, TB/Hep/HIV screening | | **Period 1** | Open Label | Active MAS825 treatment initiation, safety, and

initial efficacy assessment | | **Period 2** | Randomized | Randomized withdrawal (MAS825 vs. Placebo), double-blind disease flare monitoring | | **Period 3/3s** | Follow-up | Open-label long-term safety follow-up and sustained efficacy/survival monitoring |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Clinical efficacy, safety, and tolerability, primarily measured by the **time to disease flare** during the randomized-withdrawal period (Period 2). **Measurement Method:** Clinical assessment of flare criteria (including Physician Global Assessment [PGA], and laboratory values such as CRP, Ferritin, and systemic IL-18 levels).

7.2 Secondary & Exploratory Endpoints 1. Long-term safety and tolerability assessment through the tracking of Adverse Events (AEs) and SAEs. 2. Pharmacokinetics (PK) and immunogenicity profiles of MAS825. 3. Changes from baseline in inflammatory biomarker parameters (e.g., CRP, ferritin, cytokines).

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring across the multi-year study period for treatment-emergent adverse events (TEAEs), paying specific attention to infection risks associated with IL-1β/IL-18 blockade. **Serious Adverse Events (SAEs):** Safety events requiring hospitalization, severe infections, or events resulting in persistent disability must be reported within standard 24-hour regulatory timelines. **Data Monitoring Committee (DMC):** Utilized to regularly review unblinded data to ensure the safety of trial participants, given the vulnerable pediatric and rare disease population included in the trial.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) principles apply for primary efficacy outcomes during the randomized-withdrawal period; the Safety Set is used for all AE/SAE evaluations for patients receiving at least one dose. **Sample Size Justification:** Target enrollment is designed appropriately for ultra-rare monogenic IL-18 driven diseases to provide sufficient power to detect a clinically meaningful difference in the time to disease flare between the MAS825 and placebo groups. **Handling of Missing Data:** Standard survival analysis metrics (e.g., Kaplan-Meier estimates) for time-to-flare and standard imputation techniques used to handle drop-outs.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study is conducted in accordance with Good Clinical Practice (GCP) guidelines, the Declaration of Helsinki, and relevant regulatory requirements. **Monitoring Plan:** Routine remote and onsite monitoring visits to confirm adherence to the complex multi-period study design, verification of genetic testing, and proper blinding procedures. **Audit Trail:** Comprehensive electronic Case Report Forms (eCRFs) are maintained with full audit capabilities to ensure data integrity during the randomized withdrawal and safety extension phases.

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11. Study Identifiers & Governance

Primary ID: CMAS825D12201 **Registry Number:** NCT04641442 **Sponsor/ Lead Organization:** Novartis Pharmaceuticals **Study Title:** MASter-1 PoC study **Official Regulatory Title:** A three-period multicenter study, with a randomized-withdrawal, double-blinded, placebo-controlled design to evaluate the clinical efficacy, safety and tolerability of MAS825 in patients with monogenic IL-18 driven autoinflammatory diseases, including NLRC4-GOF, XIAP deficiency, or CDC42 mutations

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12. Clinical Framework (Autoinflammatory Specific)

Primary Condition: Monogenic IL-18 driven autoinflammatory diseases (NLRC4-GOF, XIAP Deficiency, CDC42 Mutations) **Diagnosis Threshold:** Genetic confirmation of the specific mutation alongside active clinical signs of autoinflammation **Medical Methodology:** Bispecific cytokine blockade (IL-1β and IL-18) **Optimization Target:** Prevention of autoinflammatory disease flares and avoidance of secondary macrophage activation syndrome (MAS)

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13. Study Procedures & Methodology

Management Pathway: Open-label disease stabilization followed by a double-blinded withdrawal paradigm to firmly establish therapeutic dependence **Education Protocol (HCP):** Site training focused on strict definitions of disease flare, safe administration of biologic therapy, and rigorous infection surveillance protocols. **Education Protocol (Patient):** Counseling to caregivers and patients regarding the signs of systemic flares, the risks of immunosuppression, and adherence to the prohibition of live vaccines. **Quality Audit Mechanism:** Central review of genetic eligibility and standardization of clinical flare assessments across global sites.

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14. Observation & Follow-Up Schedule

Total Duration: Approximately 4 to 5 years (combining Periods 1, 2, 3, and 3s) **Onsite Visit Cadence:** Regular scheduled visits strictly maintained during Period 1 and heavily monitored during the high-risk randomized-withdrawal window (Period 2). **Remote Monitoring Frequency:** Supplemental virtual contacts configured to ensure safety and symptom tracking between clinic visits. **Checklist Review Interval:** Routine continuous lab reviews (e.g., LFTs, CRP, Ferritin, CBC) structured to preemptively identify MAS or impending systemic flares.

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	**[DD-MMM-YYYY]**
Clinical Operations Lead	**[Name]**	___	***[DD-MMM-YYYY]***
Lead Statistician	***[Name]***	_____	[DD-MMM-YYYY]